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END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech.

Semester: I

Name of the Paper: Engineering Mathematics-I

Paper Code: TMA-101

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1		(20 marks)	CO1
a)	Find the rank of the following matrix: $A = \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{bmatrix}$ where a, b, c being all real.		
b)	For what values of k , the equations $x + y + z = 1$; $2x + y + 4z = k$; $4x + y + 10z = k^2$ have a solution and solve them completely in each case.		
c)	Diagonalise the matrix $A = \begin{bmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{bmatrix}$ by similarity transformation.		
Q2		(20 marks)	CO2
a)	Define the term 'indeterminate form' in the context of limits and evaluate the following limit $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2 \sin x}{x - \sin x}$ by clearly indicating the indeterminate form at each step where L'Hôpital's Rule is used.		
b)	If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, prove that $y_n = (n - 1)^2 y_{n-2}$ for $x = 0$.		
c)	Obtain the fourth degree polynomial approximation to the function $f(x) = e^{2x}$ about the point $x = 1$. And find maximum error when $0 \leq x \leq 0.5$		

Q3		(20 marks)	CO3
a)	If u, v, w are the roots of the cubic equation $(x - a)^3 + (x - b)^3 + (x - c)^3 = 0$, then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.		
b)	If $u = \tan^{-1} \frac{y^2}{x}$, show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\sin 2u \sin^2 u$.		
c)	Find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.		
Q4		(20 marks)	CO4
a)	Prove that $\left[m \left(m + \frac{1}{2} \right) \right] = \frac{\sqrt{\pi}}{(2)^{2m-1}} \Gamma(2m)$, where m is positive.		
b)	Use Change of order of integration to evaluate $\int_0^a \int_{y^2/a}^y \frac{y}{(a-x)\sqrt{ax-y^2}} dx dy$		
c)	Find the volume of the cylindrical column standing on the area common to the parabolas $x = y^2, y = x^2$ as base and cut off by the surface $z = 12 + y - x^2$.		
Q5		(20 marks)	CO5
a)	Find the directional derivative of $f(x, y, z) = e^{2x} \cos yz$ at $(0, 0, 0)$ in the direction of the tangent to the curve $x = a \sin t, y = a \cos t, z = at$ at $t = \pi/4$.		
b)	If $\vec{F} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$, evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along the following path C: the straight line from $(0, 0, 0)$ to $(1, 0, 0)$, then to $(1, 1, 0)$ and then to $(1, 1, 1)$.		
c)	Evaluate by Green's Theorem $\int_C [(\cos x \sin y - xy) dx + \sin x \cos y dy]$ where C is the circle $x^2 + y^2 = 1$.		